

Структура и функция ДНК и РНК. «Центральная догма» молекулярной биологии. Гены и геномы



Чугунов Антон

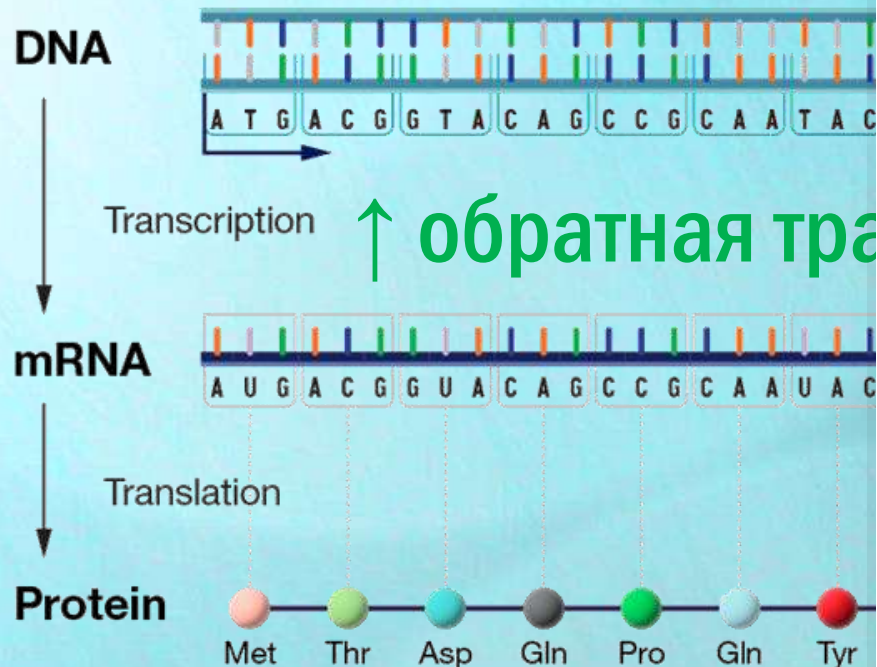
Группа анализа структуры
мембранных белков *in silico*

<https://www.ibch.ru/structure/groups/giamps>

Долгопрудный, МФТИ
7 декабря 2023 г.

Центральная догма молекулярной биологии

↻ репликация



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Central Dogma of Molecular Biology

by

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The central dogma of molecular biology deals with the detailed residue-by-residue transfer of sequential information. It states that such information cannot be transferred from protein to either protein or nucleic acid.

"The central dogma, enunciated by Crick in 1958 and the keystone of molecular biology ever since, is likely to prove a considerable over-simplification."

This quotation is taken from the beginning of an unsigned article¹ headed "Central dogma reversed", recounting the very important work of Dr Howard Temin² and others³ showing that an RNA tumour virus can use viral RNA as a template for DNA synthesis. This is not the first time that the idea of the central dogma has been misunderstood, in one way or another. In this article I explain why the term was originally introduced, its true meaning, and state why I think that, properly understood, it is still an idea of fundamental importance.

The central dogma was put forward⁴ at a period when much of what we now know in molecular genetics was not established. All we had to work on were certain fragmentary experimental results, themselves often rather uncertain and confused, and a boundless optimism that the basic concepts involved were rather simple and probably much the same in all living things. In such a situation well constructed theories can play a really useful part in stating problems clearly and thus guiding experiment.

The two central concepts which had been produced, originally without any explicit statement of the simplification being introduced, were those of sequential information and of defined alphabets. Neither of these steps was trivial. Because it was abundantly clear by that time that a protein had a well defined three dimensional structure, and that its activity depended crucially on this structure, it was necessary to put the folding-up process on one side, and postulate that, by and large, the polypeptide chain folded itself up. This temporarily reduced the central problem from a three dimensional one to a one dimensional one. It was also necessary to argue that in spite of the miscellaneous list of amino-acids found in proteins (as then given in all biochemical textbooks) some of them, such as phosphoserine, were secondary modifications; and that there was probably a universal set of twenty used throughout nature. In the same way minor modifications to the nucleic acid bases were ignored; uretic in RNA was considered to be informationally

analogous to thymine in DNA, thus giving four standard symbols for the components of nucleic acid.

The principal problem could then be stated as the formulation of the general rules for information transfer from one polymer with a defined alphabet to another. This could be compactly represented by the diagram of Fig. 1 (which was actually drawn at that time, though I am not sure that it was ever published) in which all possible simple transfers were represented by arrows. The arrows do not, of course, represent the flow of matter but the directional flow of detailed, residue-by-residue, sequence information from one polymer molecule to another.

Now if all possible transfers commonly occurred it would have been almost impossible to construct useful theories. Nevertheless, such theories were part of our everyday discussions. This was because it was being tacitly assumed that certain transfers could not occur. It occurred to me that it would be wise to state these pre-conceptions explicitly.

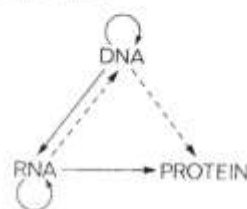


Fig. 1. The arrows show the situation as it seemed in 1958. Solid arrows represent possible transfers, dotted arrows possible transfers. The dotted arrows (compare Fig. 2) represent the transfers postulated by the central dogma. They are the three possible arrows starting from protein.

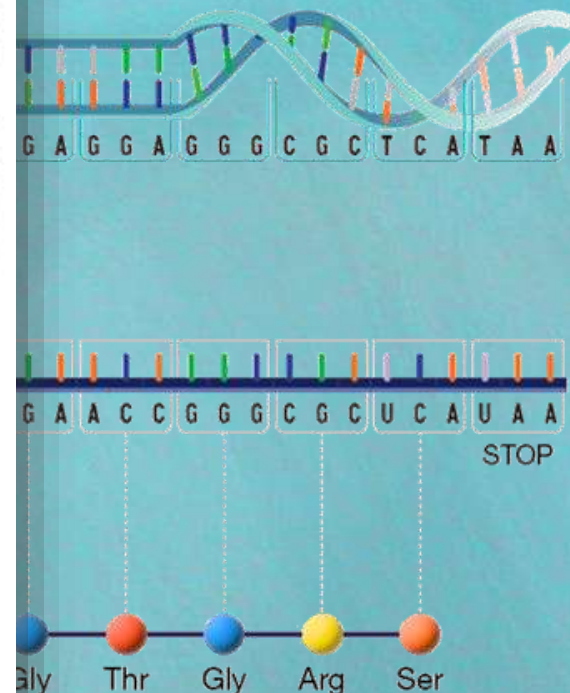
A little analysis showed that the transfer could be divided roughly into three groups. The first group was those for which some evidence, direct or indirect, seemed to exist. These are shown by the solid arrows in Fig. 2. They were:

- I (a) DNA→DNA
- I (b) DNA→RNA
- I (c) RNA→Protein
- I (d) RNA→RNA

The last of these transfers was presumed to occur because of the existence of RNA viruses.

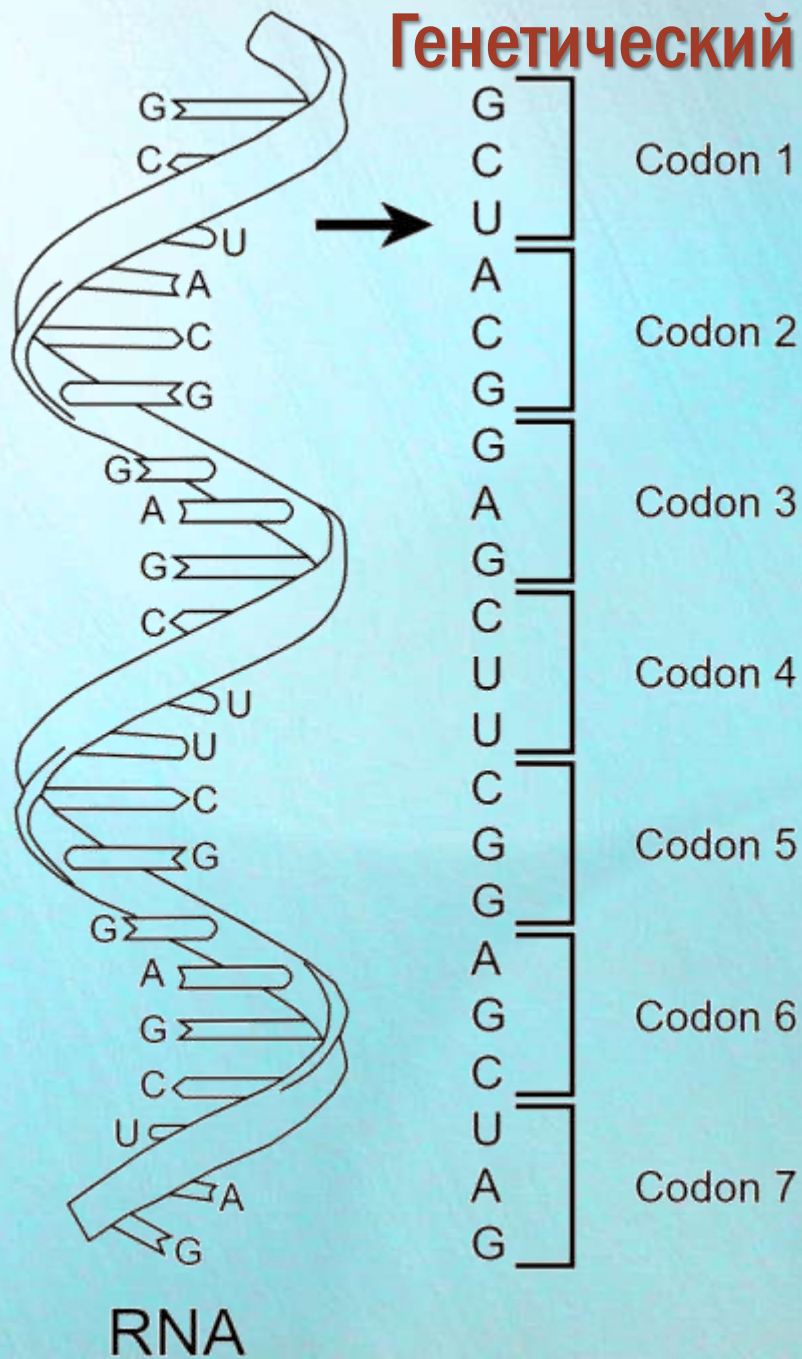
Next there were two transfers (shown in Fig. 2 as dotted arrows) for which there was neither any experimental evidence nor any strong theoretical requirement. They were:

- II (a) RNA→DNA (see the references to Temin's work²)
- II (b) DNA→Protein.

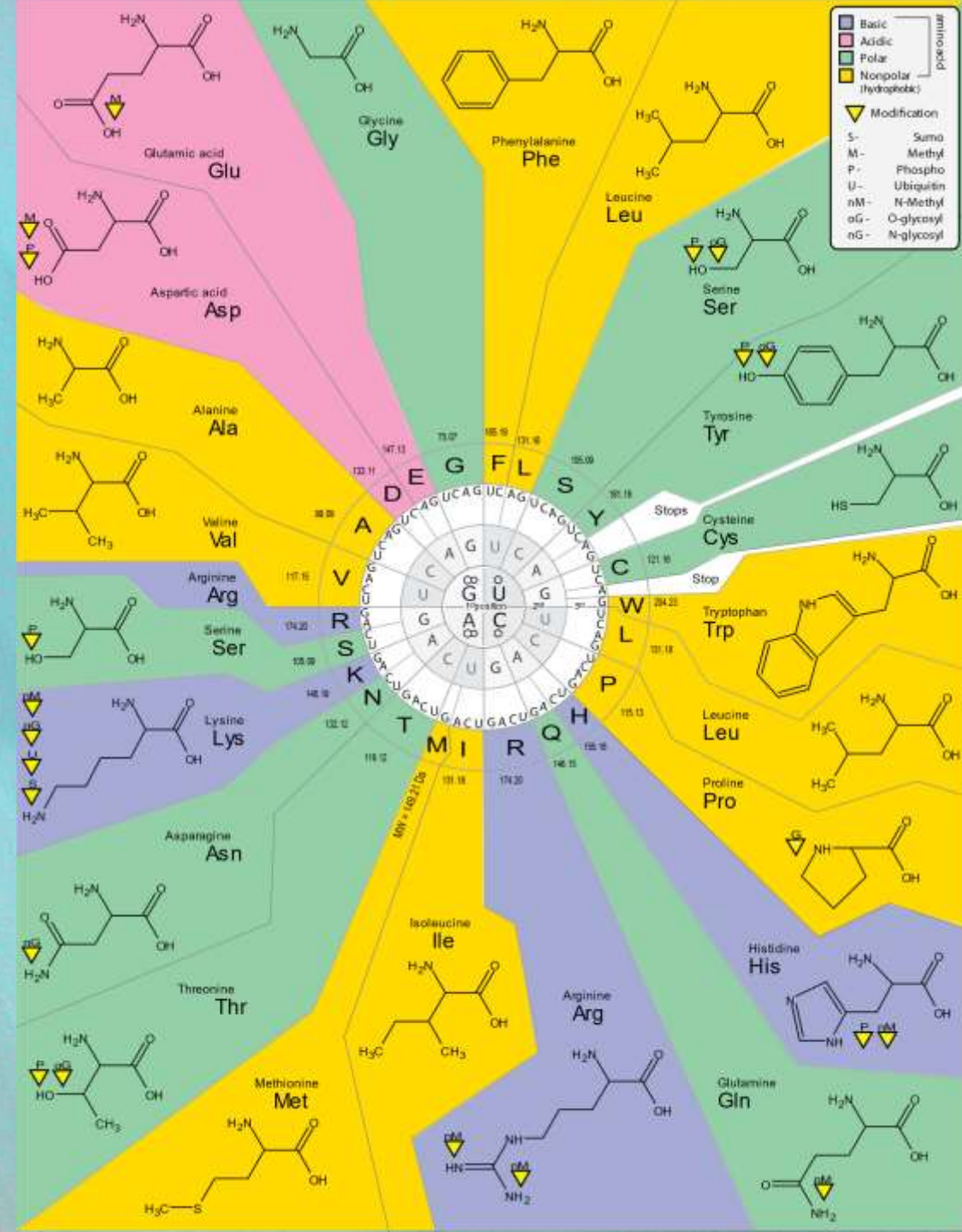


Френсис Крик, 1958, 1970

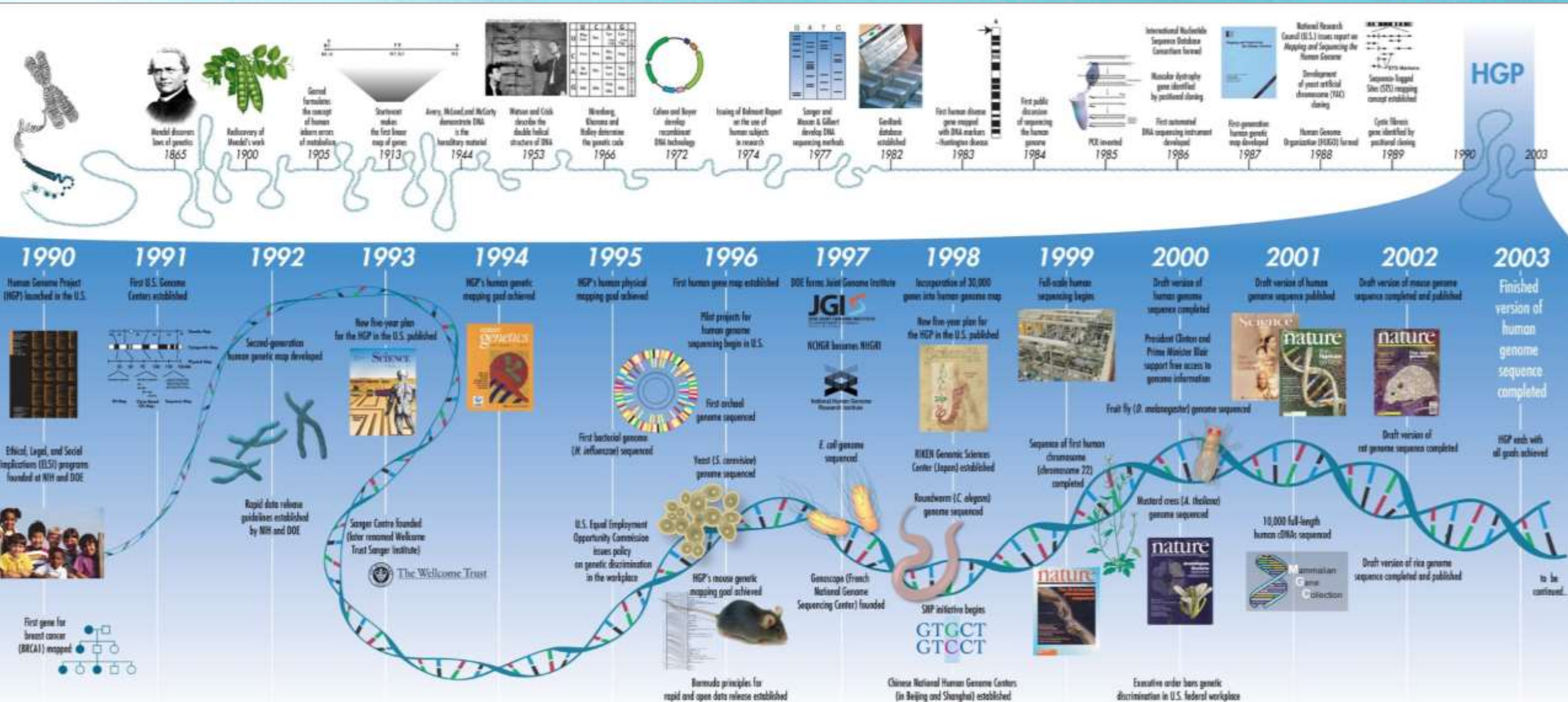
Генетический код



Wikipedia: [Genetic code](#)



Генетический таймлайн



Today in Chemistry History

25th July – Rosalind Franklin's birthday (1920) and DNA's structure



Rosalind Franklin

Born

25 July 1920

Died

16 April 1958

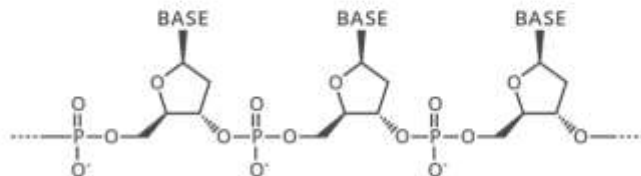


Rosalind Franklin was a chemist and X-ray crystallographer whose work was instrumental in the discovery of the structure of DNA. She missed out on a Nobel Prize for her work as they are not awarded posthumously.

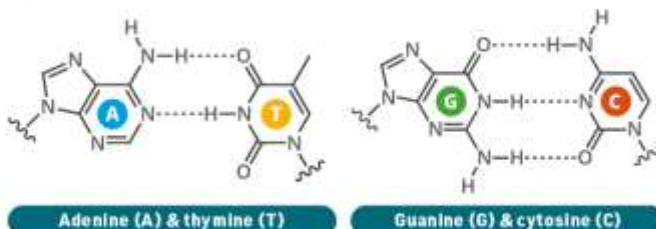
Photograph 51 and the structure of DNA



Photograph 51 is an X-ray diffraction image of DNA taken during Franklin's research. It was crucial in developing a model of DNA and confirming its double helical structure.

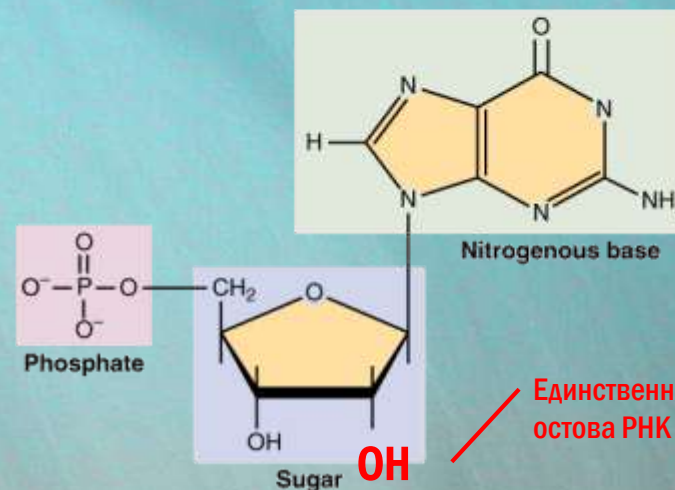
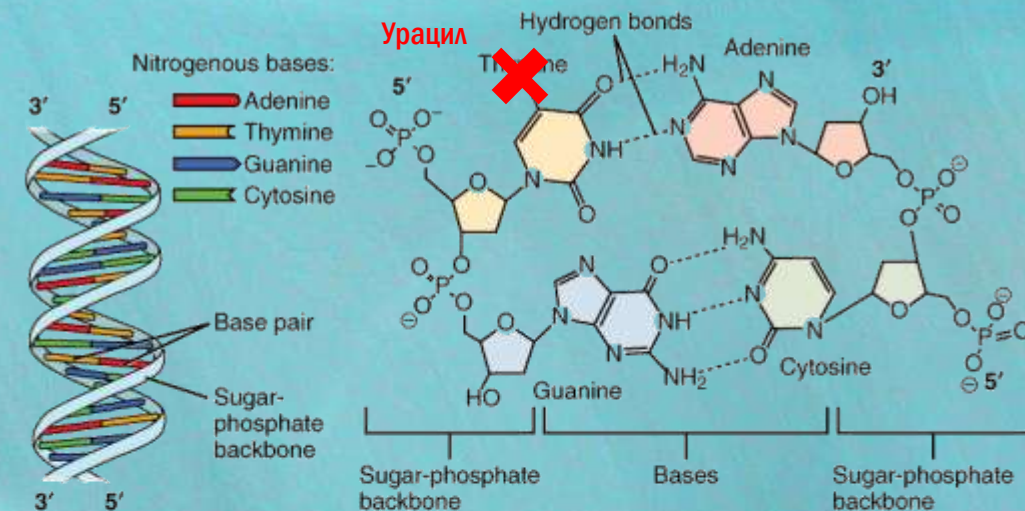


DNA is a polymer made up from monomers called nucleotides. A sugar phosphate backbone (above) forms the two strands, which are held together by hydrogen bonds between the bases found on these strands (below).



Строение нуклеиновых кислот

ДНК и РНК — полимеры, состоящие из нуклеотидов

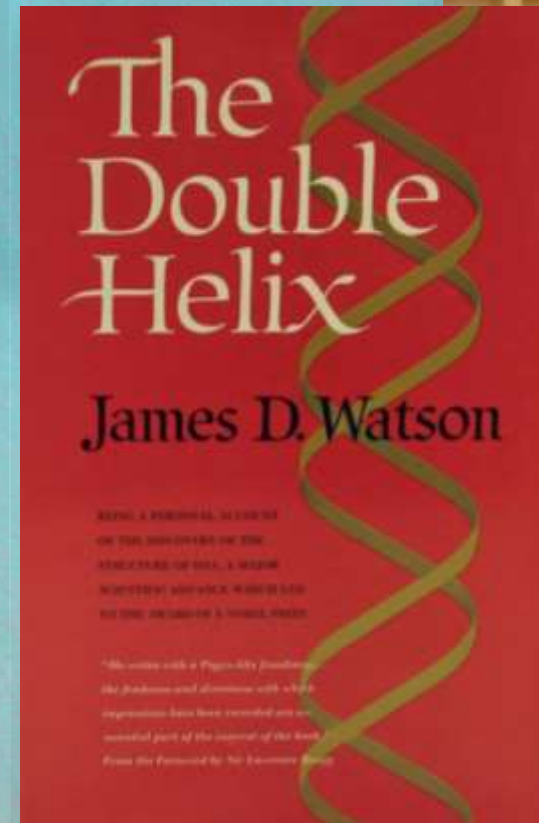


Единственное отличие
остова РНК от ДНК

Триумф двойной спирали



Морис Уилкинс, Джон Стейнбек, Джон Кендрю,
Макс Перутц, Френсис Крик, Джон Уотсон
На церемонии награждения Нобелевских премий 1962 г.

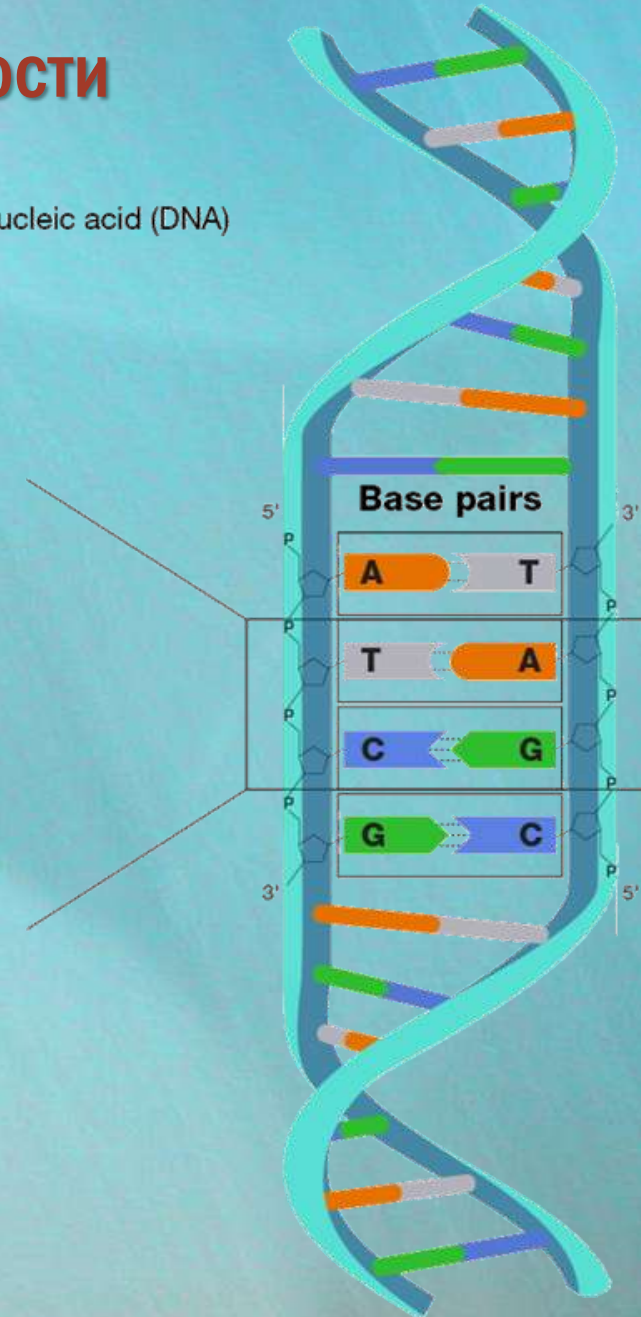
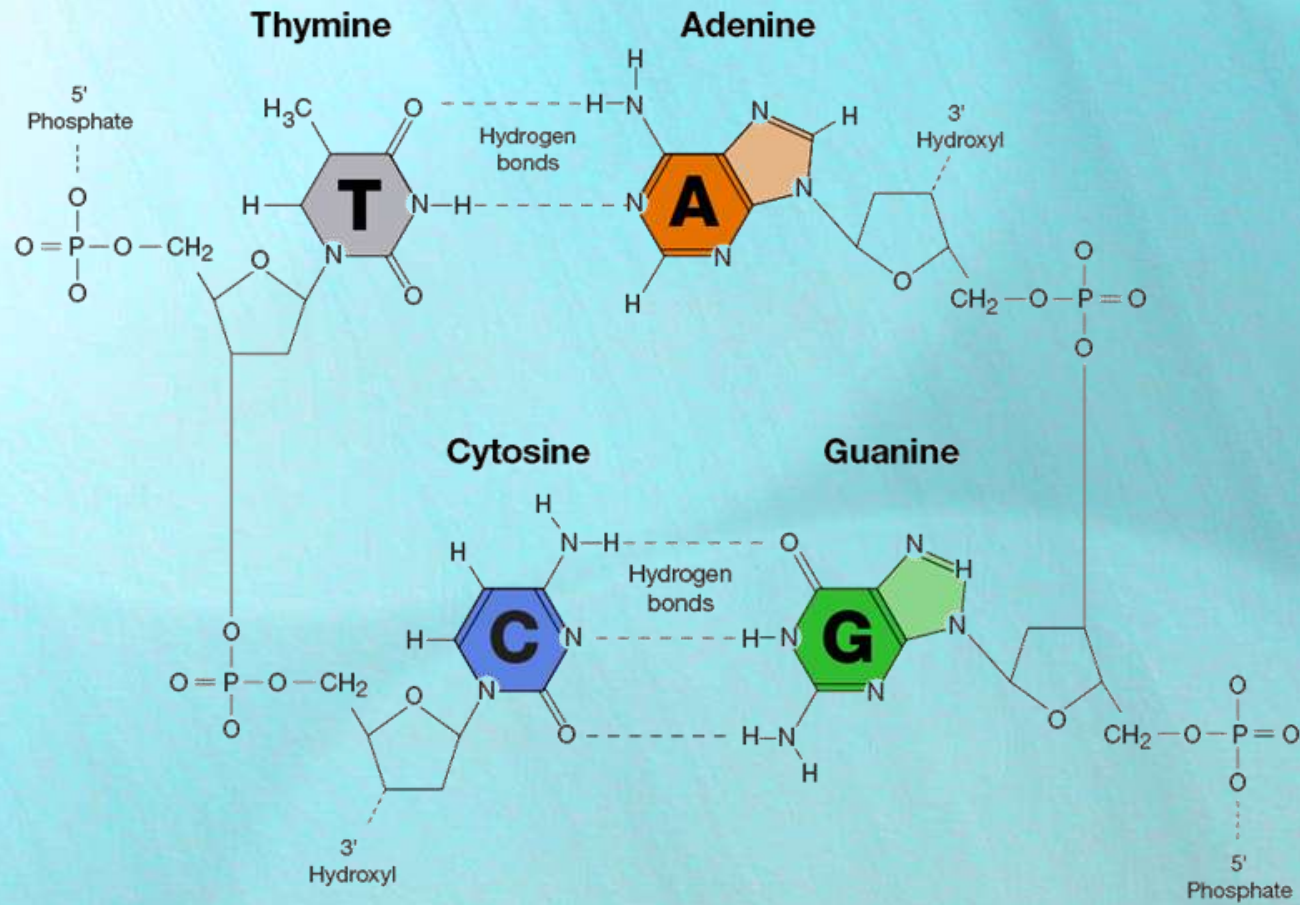


Двойная спираль

Принцип комплементарности

Base pairs

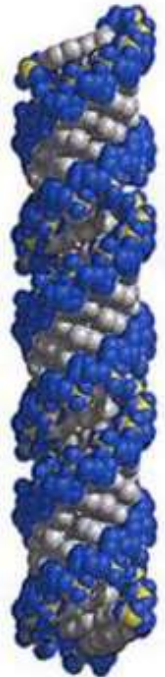
Deoxyribonucleic acid (DNA)



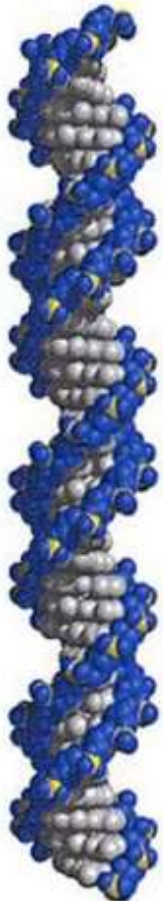


A, B, Z: такая разная двойная спираль

	A	B	Z
Helix sense	Right handed	Right-handed	Left handed
Repeating unit	1 bp	1bp	2 bp
Rotation/bp	33.6°	35.9°	60° /2
Mean bp/turn	10.7	10.0	12
Inclination of bp to axis	+19°	-1.2°	-9°
Rise/bp along axis	2.3Å	3.32Å	3.8Å
Pitch/turn of helix	24.6Å	33.2Å	45.6Å
Mean propeller twist	+18°	+16°	0°
Glycosyl angle	anti	anti	C: anti, G: syn
Sugar pucker	C3'-endo	C2'-endo	C: C2'-endo, G: C2'-exo
Diameter	26Å	20Å	18Å



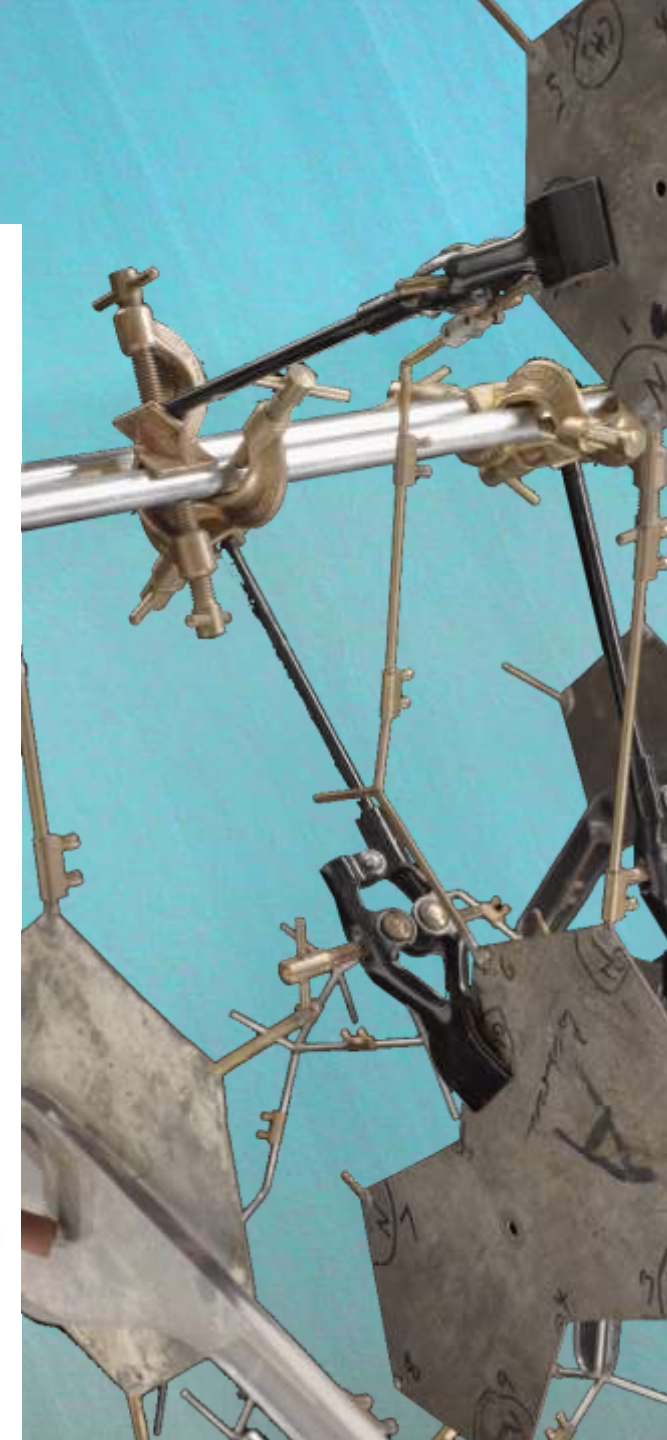
A form



B form



Z form

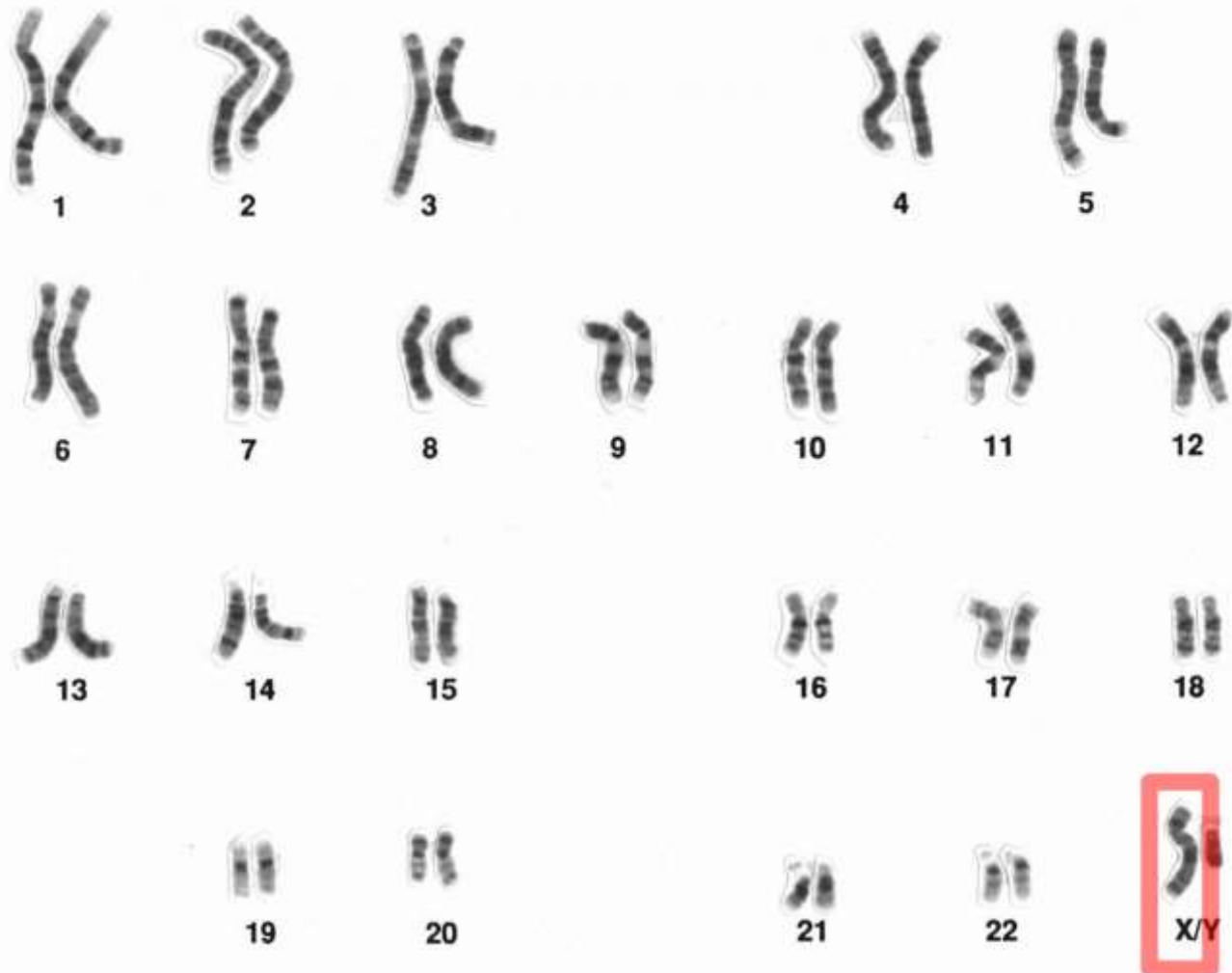


Упаковка ДНК в хромосомы

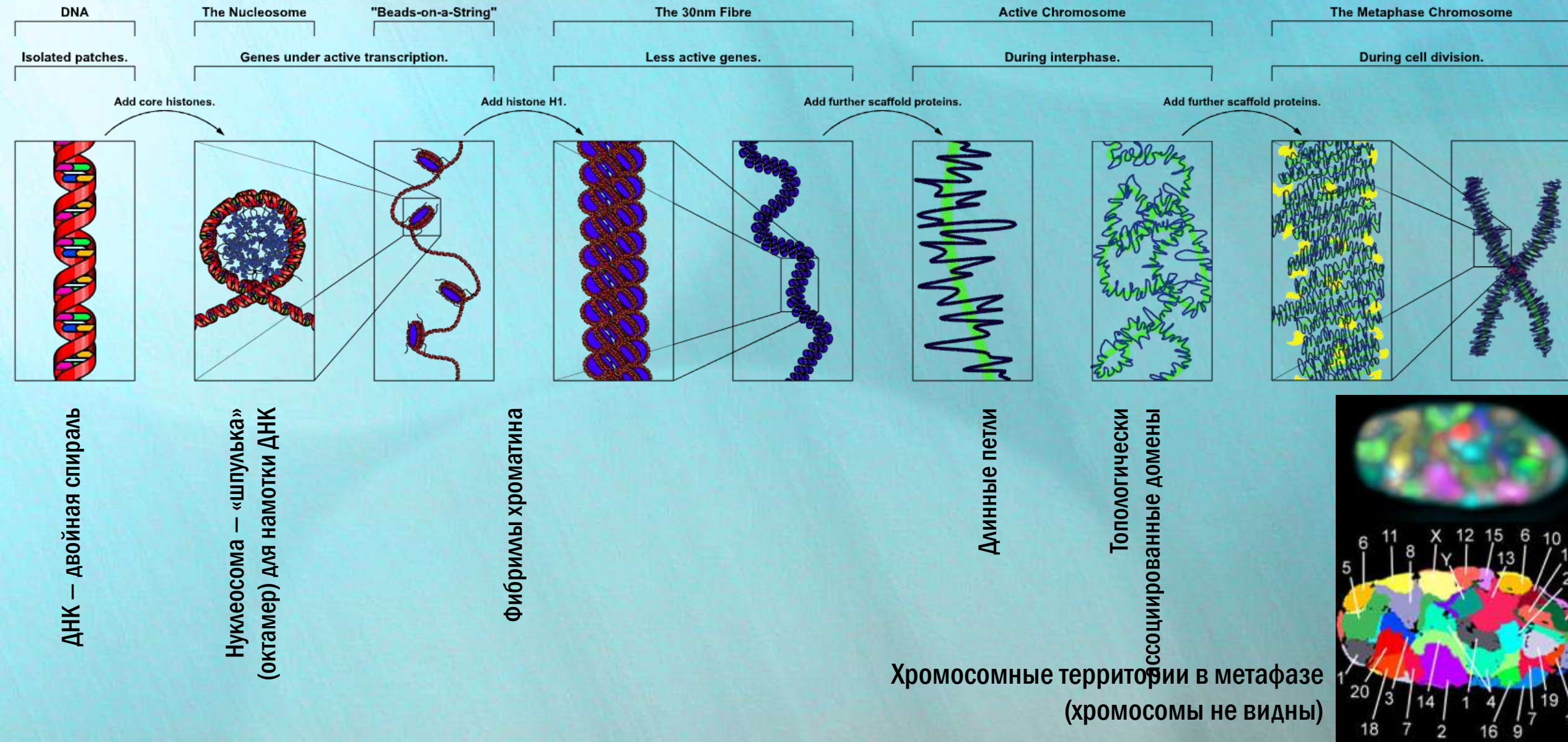
Кариотип – 23 пары хромосом человека:

$\approx 3.1 \times 10^9$ п.н.

- Геном – совокупность всего генетического материала (включая митохондриальную ДНК)
- Длина всей ДНК в клетке ≈ 2 м
- Размер клетки: ≈ 100 мкм
- Размер хромосомы < 10 мкм



Уровни упаковки хромосом



Хромосомные территории ядрышко

хромосомные
территории

интерхроматиновые
домены

ядерная
пора

ядерная
оболочка

ядерная
ламина



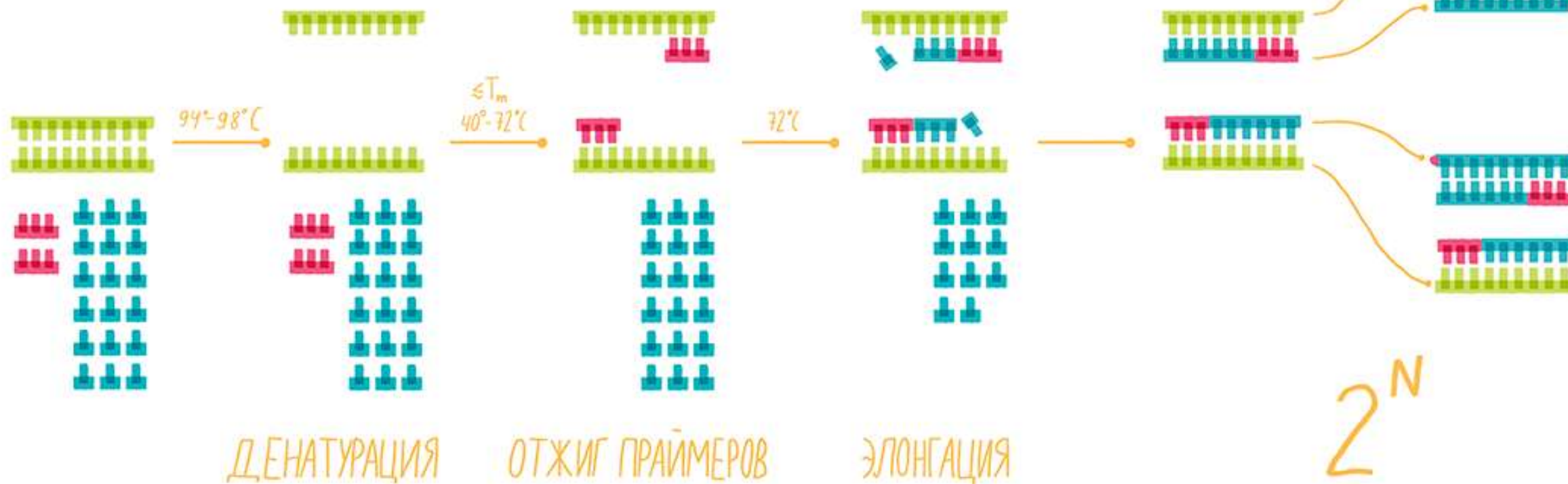
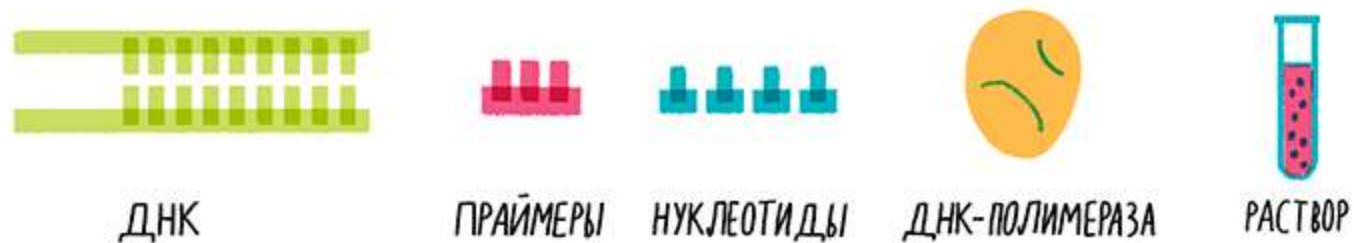
3D-организация генома

Репликация ДНК

От ДНК к белку: транскрипция и трансляция

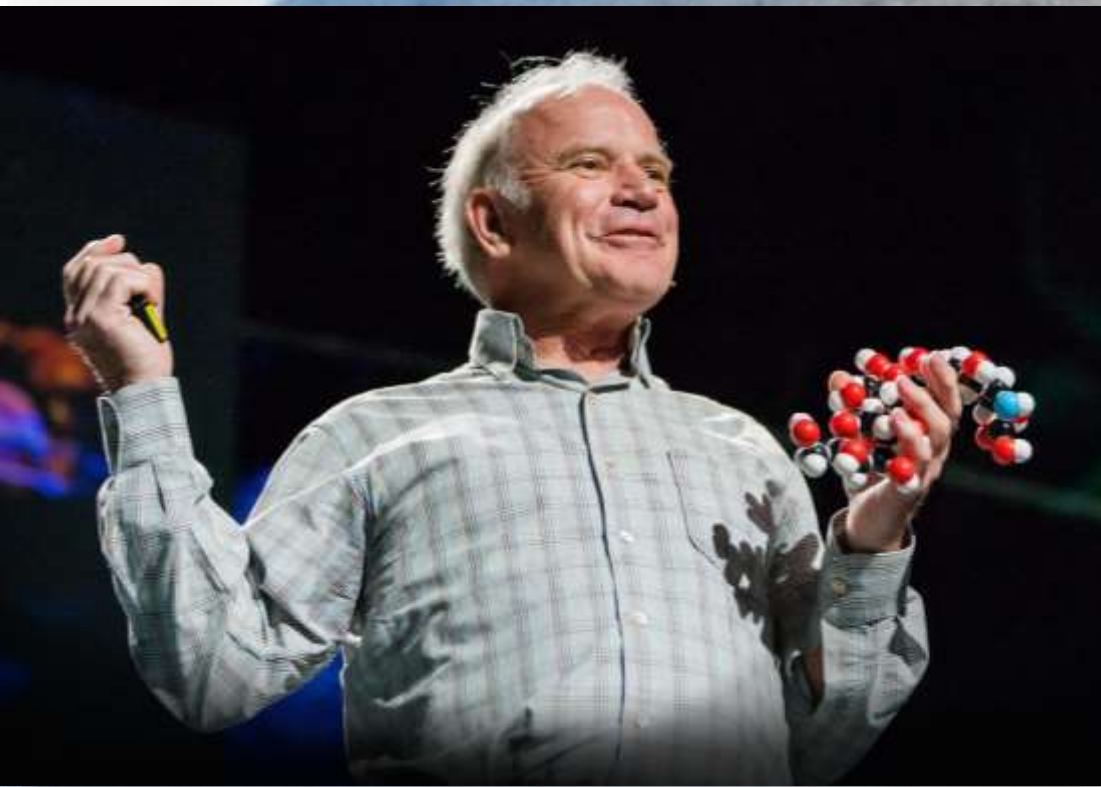


Полимеразная цепная реакция



KARY MULLIS

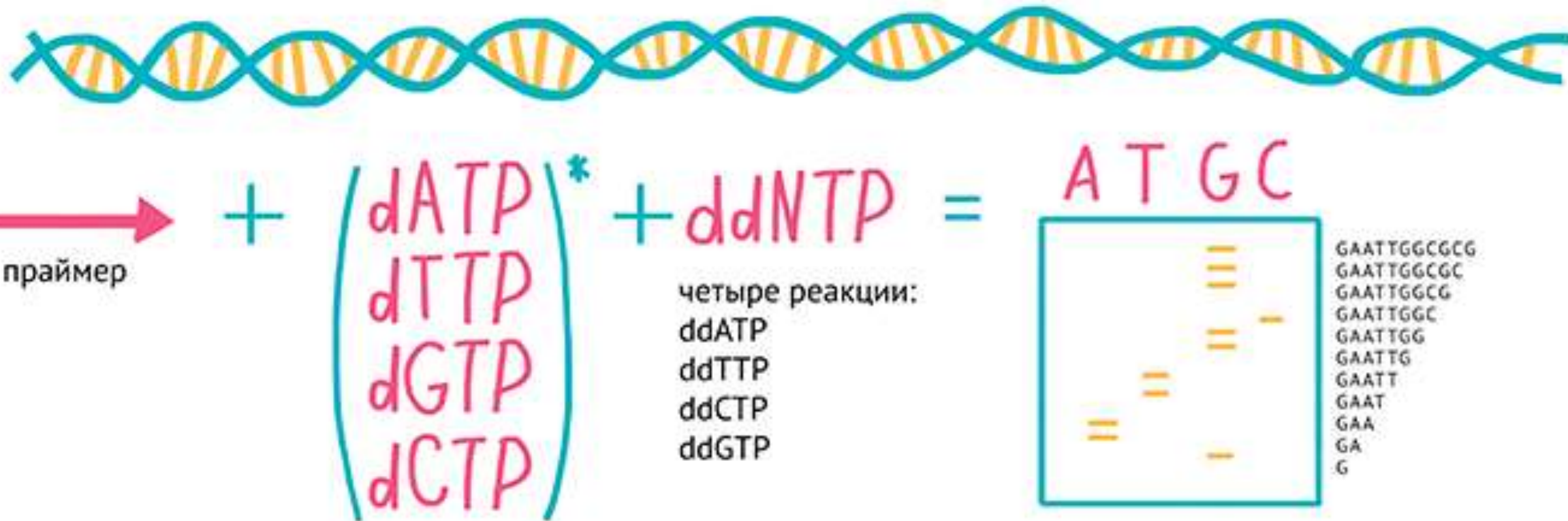
1944-2019



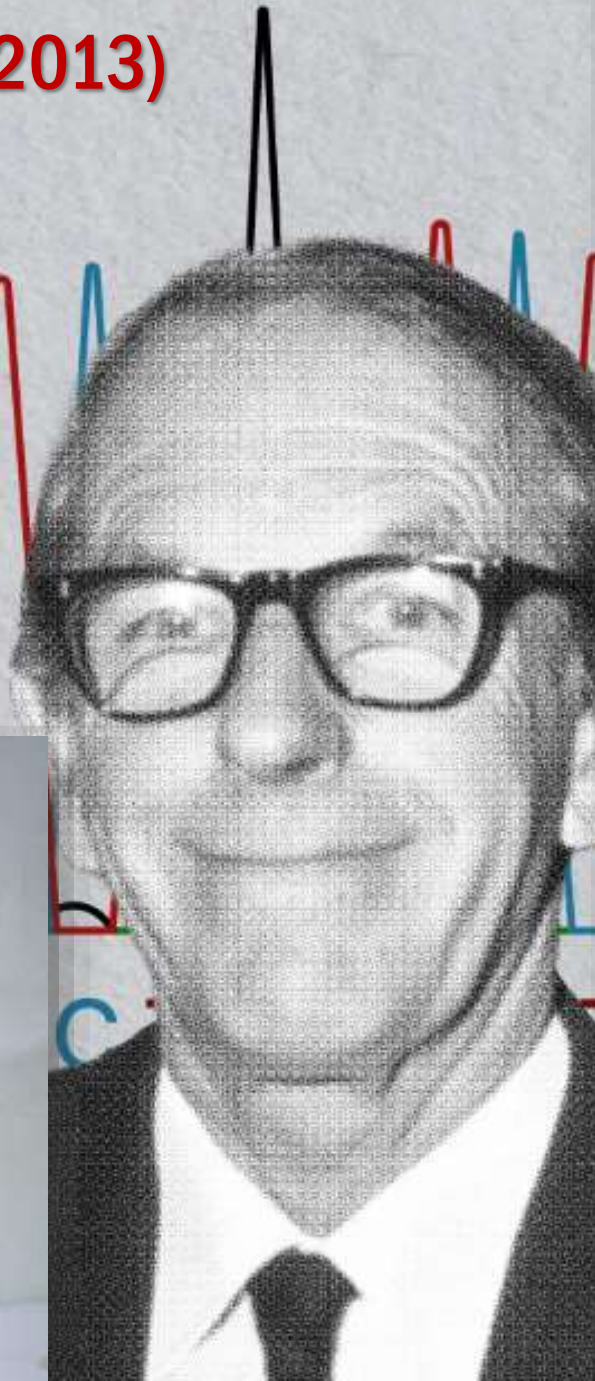
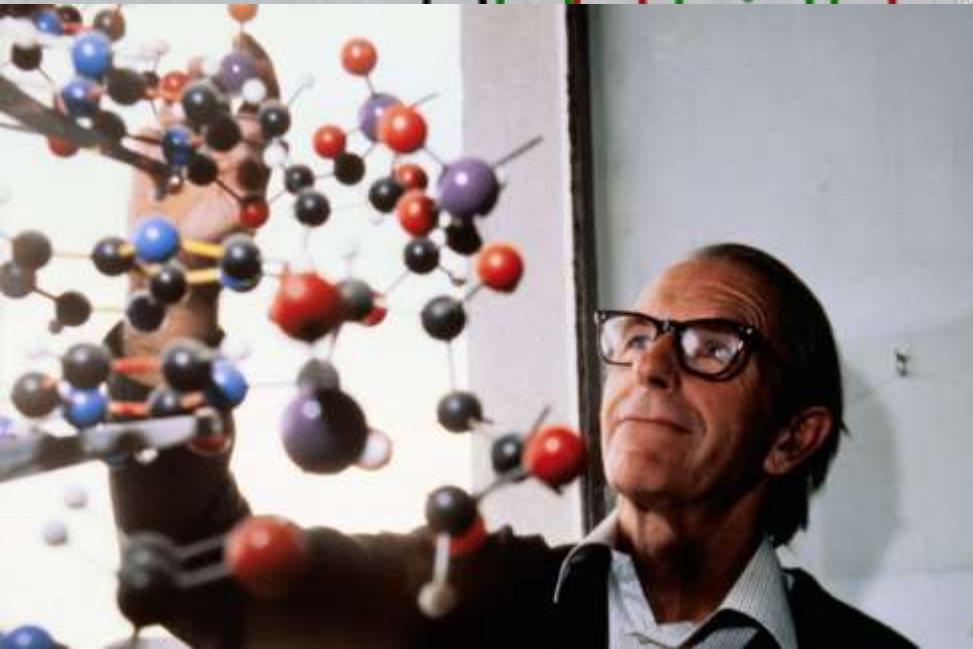
Секвенирование ДНК: метод Сенгера

Если секвенирование белков по Сенгеру устроено по принципу «разобрать», то последовательность ДНК определяется по мере матричного синтеза.

Метод «терминаторов»:



Фредерик Сенгер (1918–2013)



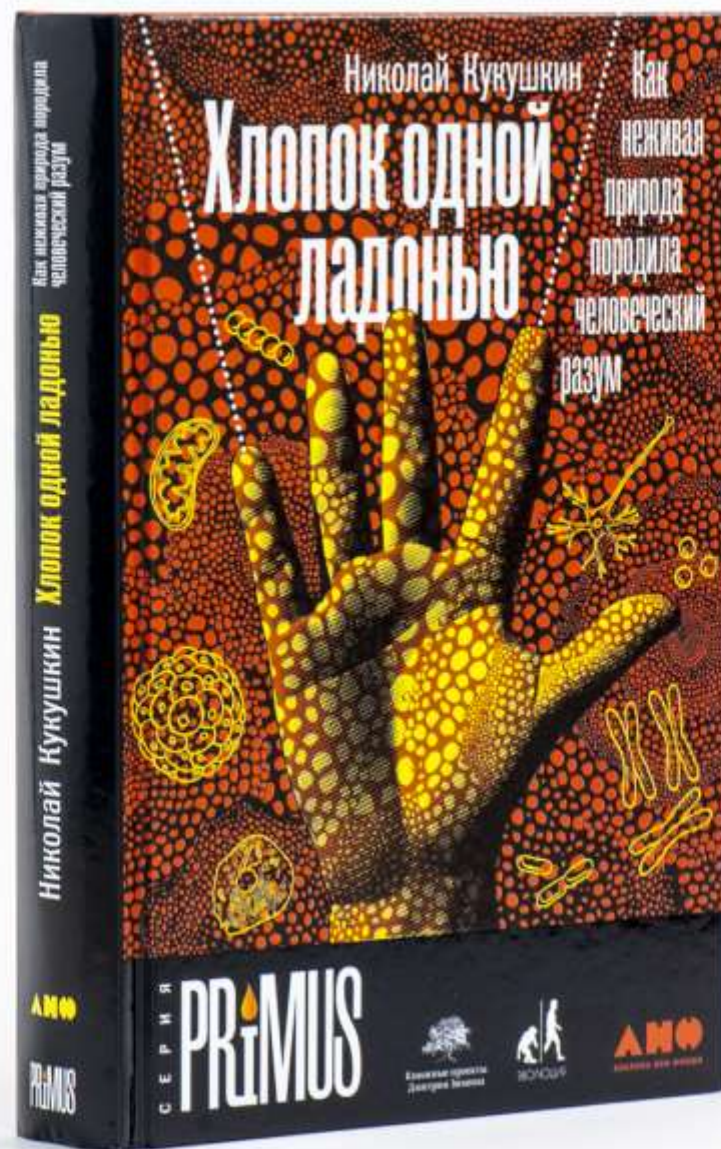
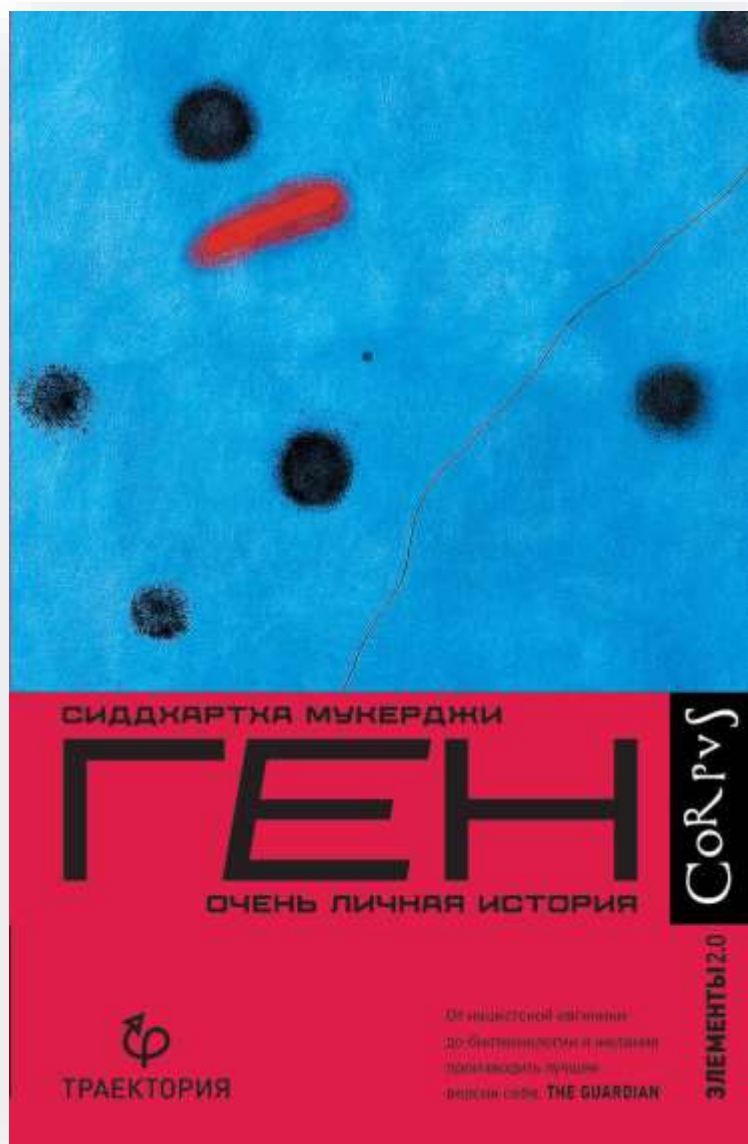
Первый человек, дважды получивший
Нобелевскую премию по химии
(и один из пяти дважды лауреатов)



- Фрагментация (shotgun)
- Регистрация включения нуклеотидов
- Биоинформатическая «сборка»



Дополнительная литература



БМ

Спасибо за внимание!



Институт биоорганической химии РАН — <http://ibch.ru>

Группа анализа структуры мембранных белков *in silico*